Arista Access Node ELEC-ACC-01 Optical Transport Failure SOP for ALARM_5312 Warning Alarm

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5312

Description

The failure of the Optical Transport system in the Arista Access Node ELEC-ACC-01 at the Electronic City Hub site occurs due to the degradation of the SFP/XFP modules or fiber cuts that result in Rx/Tx power loss. This failure is specific to the Access role in the given vendor equipment due to its reliance on the Optical Transport system to transmit data over fiber optic cables. The Optical Transport system in the Arista Access Node ELEC-ACC-01 utilizes the SFP/XFP modules to transmit and receive data over the fiber optic cables. When the SFP/XFP modules degrade or the fiber optic cables are cut, the Rx/Tx power loss occurs, resulting in a failure of the Optical Transport system. The Alarm Code ALARM_5312 is triggered when the system detects a warning condition, indicating a potential issue with the Optical Transport system that requires immediate attention to prevent further disruptions to network services.

Pre-requisite Checks

- Verify that the Electronic City Hub site is accessible and that all necessary equipment is in place.

- Check the status of the redundant link to ensure that it is functioning properly and can take over in case of a failure.

- Dispatch remote hands to the Electronic City Hub site to assist in the diagnosis and resolution of the issue.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP module	High
Fiber cut or damage	Medium
Rx/Tx power loss due to software or configuration issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface status: Up, Link state: Up, Description: This command is used to verify the status of the interface and ensure that it is up and functioning properly.

Command: show optics, Expected Output: Optics status: Normal, Temperature: 25C, Description: This command is used to verify the status of the optics and ensure that they are functioning properly and within normal temperature ranges.
```

```
Command: show tx-power, Expected Output: Tx power: 0 dBm, Description: This command is used to verify the transmit power of the interface and ensure that it is within normal ranges.
```

Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP module, Command: replace optics slot 1 with new optics, Description: Replace the degraded SFP/XFP module with a new one to restore the Optical Transport system to normal functioning.

Step 2: Step: Verify the status of the optics, Command: show optics, Description: Verify that the optics are functioning properly and within normal temperature ranges.

Step 3: Step: Verify the transmit power, Command: show tx-power, Description: Verify that the transmit power is within normal ranges.

Post-Check Verification

```
Command: show interface status, Expected Output: Interface status: Up, Link state: Up, Description: Verify that the interface is up and functioning properly.

Command: show optics, Expected Output: Optics status: Normal, Temperature: 25C, Description: Verify that the optics are functioning properly and within normal temperature ranges.

Command: show tx-power, Expected Output: Tx power: 0 dBm, Description: Verify that the transmit power is within normal ranges.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace optics slot 1 with original optics, Description: Restore the original SFP/XFP module to its original position to ensure that the Optical Transport system is restored to its original state.

- Step: Verify the status of the optics, Command: show optics, Description: Verify that the optics are functioning properly and within normal temperature ranges.

- Step: Verify the transmit power, Command: show tx-power, Description: Verify that the transmit power is within normal ranges.